C 80923 (Pages : 3) Name.....

Reg. No.....

FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2020

Physics/Applied Physics

PHY 4B 04/APY 4B 04—ELECTRODYNAMICS—I

Time : Three Hours Maximum : 80 Marks

Section A

Answer **all** *questions.*
Each question carries 1 *mark.*
Answer in a word **or** *phrase.*

1. State Gauss's law in differential form. Explain the symbols used.
2. Give the relation between vector potential and magnetic field.
3. If the electrostatic potential in a region is represented as $V = 2x + 3y - z$, obtain the expression for electric field strength.
4. Write down the wave equation in one dimension.
5. What are equipotential surfaces ?

State whether the statement is True *or* False :

6. If the electric field is zero at a point, the electric potential is necessarily zero at that point.
7. The number of electric field lines penetrating normal to a surface is called electric flux.
8. Above the Curie temperature, a ferromagnetic material would become paramagnetic.
9. CO_2 is an example of non polar molecule.
10. In the interior of a dielectric kept in a uniform field, the net charge is zero.

(10 × 1 = 10 marks)

Section B

Answer **all** *questions in two* **or** *three sentences.*
Each question carries 2 *marks.*

11. Show that electrostatic energy does not obey superposition principle.
12. Define one ampere current using the expression for the force acting per unit length of long parallel current carrying conductors.
13. Find the vector potential of an infinite solenoid with n turns per unit length, radius R and current I.
14. Is dielectric an insulator ? Explain.

15. What is meant by dielectric breakdown ?

16. Distinguish between permeability and permittivity.

17. Write the relation connecting M, B and H.

(7 × 2 = 14 marks)

Section C

Answer any **five** *questions.*
Each question carries 4 *marks*

18. Explain scalar potential and vector potential.

19. Derive the expression for the capacitance of two concentric metal shells with radii α and β.

20. Explain cyclotron motion and cyclotron frequency.

21. Explain electrostatic boundary conditions.

22. Derive the Poisson's equation and Laplace's equation.

23. Derive an expression for the energy of a point charge distribution.

24. Write Ampere's law in differential and integral form.

(5 × 4 = 20 marks)

Section D

Answer any **four** *questions.*
Each question carries 4 *marks*

25. A metallic sphere of radius 10 cm. has a surface charge density of 10 nC/m^2. Calculate the energy stored in the system.

26. A current distribution gives rise to the magnetic vector potential $A = x^2y\ \hat{x} + y^2x\ \hat{y} - 4xyz\ \hat{z}$. Calculate B at (– 1, 2, 5).

27. A 50 cm. long solenoid having 500 turns and radius 2 cm. is wound on an iron core of relative permeability 800. What will be the average e.m.f. induced in the solenoid if the current in it changes from 0 to 2A in 0.05 *s* ?

28. Two point charges – 4 μC and 5 μC are located at (2, – 1, 3) and (0, 4, – 2). Find the potential at (1, 0, 1) assuming zero potential at infinity.

29. A capacitor is constructed from two conductive metal plates 30 cm. × 50 cm. which are spaced 6 mm. apart from each other, and uses dry air as its only dielectric material. Calculate the capacitance of the capacitor.

30. The electric field in some region of space is found to be $E = kr^3 \hat{r}$ in spherical Co-ordinates, where k is some constant and $\hat{r}$ is the unit vector. Find the charge density.

31. In an electric field, the potential at any point is given by $V = 5x^2 - 3y^2 + 2z^2$ volts. Find the expression for electric field intensity at a point (2,3,1). Also find its magnitude.

(4 × 4 = 16 marks)

Section E

Answer any **two** *questions.*
Each question carries 10 *marks.*

32. Show that the potential due to a polarized dielectric is the same as that produced by a volume charge density and surface charge density.

33. Use Gauss's law to find the electric field:

    (a) Outside a uniformly charged solid sphere of radius R and total charge q.

    (b) At a distance from an infinite plane of uniform surface charge density.

34. Explain the following types of magnetic materials : (a) Paramagnets ; (b) Diamagnets ; and (c) Ferromagnets.

35. (a) State and explain Biot Savart's law.

    (b) Derive an expression for the magnetic field due to a circular loop of current at a point on the axis of the coil.

(2 × 10 = 20 marks)